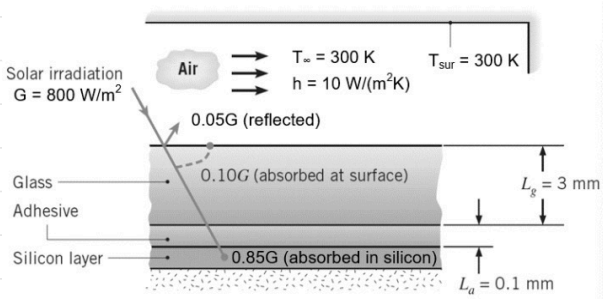
Aloyse Maille - aam299 HEAT TRANSFER

Problem Set

1. KNOWN: $L_g = 3\,mm$ $k_g = 1.5\,W/m\cdot K$ $L_a = 0.1\,mm$ $k_a = 1\,W/m\cdot K$
$\eta = a - bT_{si}$ where $a = 0.553$ and $b = 0.001\,K^{-1}$. $R_c'' = 10^{-4}\,m^2\cdot K/W$. $G = 800\,W/m^2$
5% of solar irradiation is reflected from top surface of glass, 10% absorbed, irradiation absorbed at the silicon layer, 85% transmitted/absorbed by silicon layer. $T_\infty = 300K$ $h = 10\,W/m^2\cdot K$ $T_{sur} = T_\infty = 300K$
$h_{rad} = 5\,W/m^2\cdot K$

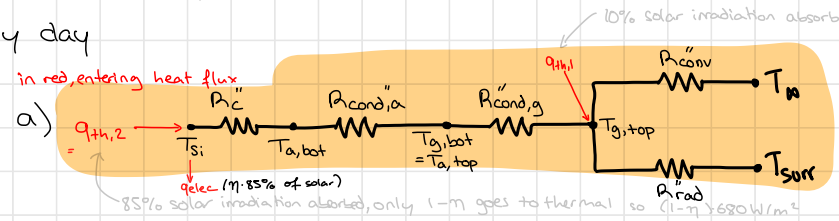
FIND: a) thermal resistance network describing HT across PV panel. state key assumptions making it valid. calculate value of each th R.
b) $T_{g,top}$ and T_{si}. c) η and resulting elec power flux produced by PV panel

SCHEMATIC:



ASSUMPTIONS: i) layer of silicon is very thin ii) backside of silicon layer is thermally insulated. iii) no contact thermal resistance between adhesive and glass. iv) sunny day

ANALYSIS: a)



key assumptions: steady-state, 1D heat conduction, uniform properties

$$R_{cond,a} = \frac{L_a}{k_a A} \qquad R_{cond,g} = \frac{L_g}{k_g A} \qquad R_{conv} = \frac{1}{hA} \qquad R_{rad} = \frac{1}{h_{rad} A}$$

So the thermal resistance per unit area are:

$$R''_c = 10^{-4}\ m^2 \cdot K/W$$

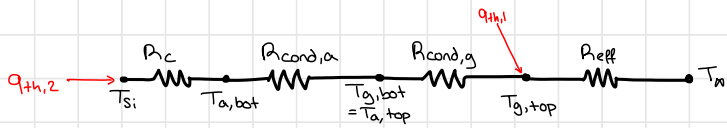
$$R''_{cond,a} = \frac{L_a}{k_a} = \frac{0.1 \cdot 10^{-3}}{1} = 10^{-4}\ m^2 \cdot K/W$$

$$R''_{cond,g} = \frac{L_g}{k_g} = \frac{3 \cdot 10^{-3}}{1.5} = 2 \cdot 10^{-3}\ m^2 \cdot K/W$$

$$R''_{conv} = \frac{1}{h} = \frac{1}{10} = 10^{-1}\ m^2 \cdot K/W$$

$$R''_{rad} = \frac{1}{h_{rad}} = \frac{1}{5} = 2 \cdot 10^{-1}\ m^2 \cdot K/W$$

b)



$$\frac{1}{R_{eff}} = \frac{1}{R_{rad}} + \frac{1}{R_{conv}} = \frac{1}{2 \cdot 10^{-1}} + \frac{1}{10^{-1}} = 15\ W/(m^2 \cdot K)$$

Energy balance over the whole network gives:

$$q_{th,1} + q_{th,2} = \frac{T_{g,top} - T_\infty}{R_{eff}} \iff 80 + (1-\eta)680 = 15 \cdot (T_{g,top} - 300) \quad (1)$$

Also can write $\dfrac{T_{si} - T_{g,top}}{R_c + R_{cond,a} + R_{cond,}} = q_{th,2} = (1-\eta)680 = \dfrac{T_{si} - T_{g,top}}{2.2 \times 10^{-3}}$

$$\iff T_{si} = T_{g,top} + 1.496(1-\eta) \quad \text{and} \quad \eta = 0.553 - 0.001 T_{si}$$

$$\text{so} \quad \eta = \frac{0.5515 - 0.001 T_{g,top}}{0.998504} \simeq 0.55 - 0.001 T_{g,top}$$

Plugging (2) in (1), $5260 - 680\eta = 15 T_{g,top} \Rightarrow 5260 - 374 + 0.68 T_{g,top} = 15 T_{g,top}$

so $T_{g,top} = 341.2\ K$ so $\eta = 0.21$ and $T_{si} = 343\ K$

c) Again we have $\eta = 0.553 - 0.001 \cdot (343) = 0.21$

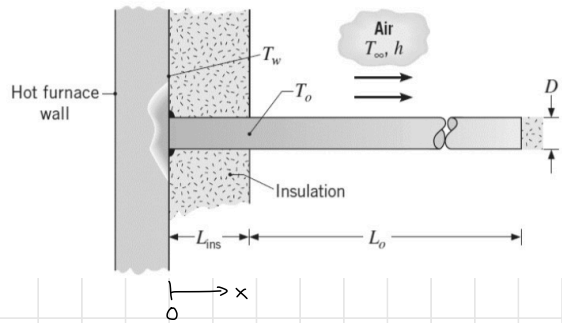
85% of $800 W/m^2$

and $q_{elec} = \eta \cdot 680 = 142.8 W/m^2$

2. **KNOWN**: $D = 10\,mm$ $k = 80\,W/m\cdot K$ $T_w = 200°C$ $L_{ins} = 200\,mm$
need $T_o < T_{max} = 100°C$ $T_\infty = 20°C$ $h = 20\,W/m^2\cdot K$

FIND: a) T_o as function of thermal geometrical parameters.
b) If $L_o = 200\,mm$, calculate T_o. Does it meet the specified operating limit? c) Minimum L_o ensuring $T_o < T_{max}$.

SCHEMATIC:



ASSUMPTIONS: i) rod tip is well insulated ii) 1D heat conduction iii) steady-state iv) uniform properties

ANALYSIS: a) No convection, only conduction from wall to $x = L_{ins}$. in that section,

$$q_{ins} = -kA\frac{\Delta T}{L} = -k\pi\left(\frac{D}{2}\right)^2 \frac{T_o - T_w}{L_{ins}}$$

for the rest of the rod, tip is insulated so adiabatic and

$$q_f = M\tanh(mL_o) \quad \text{with} \quad M = \sqrt{hPkA_c}\,\theta_o = \sqrt{h\,2\pi\frac{D}{2}\,k\pi\left(\frac{D}{2}\right)^2}\,\theta_o = \frac{\pi D}{2}\sqrt{hkD}\,\theta_o$$

with $\theta_o = T_o - T_\infty$

$$\text{and } m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{h\,2\pi\,D/2}{k\,\pi(D/2)^2}} = \sqrt{\frac{4h}{kD}}$$

so at $x = L_{ins}$, the energy balance gives:

$$-k\pi\frac{D^2}{4}\frac{T_o - T_w}{L_{ins}} = \frac{\pi D}{2}\sqrt{hkD}\,\theta_o\tanh\left(\sqrt{\frac{4h}{kD}}\cdot L_o\right)$$

$$\Leftrightarrow \quad \frac{T_o - T_w}{L_{ins}} = -\underbrace{\sqrt{\frac{4h}{kD}}}_{m}(T_o - T_\infty)\tanh\left(\underbrace{\sqrt{\frac{4h}{kD}}}_{m}\cdot L_o\right)$$

$$\Leftrightarrow \frac{T_0}{L_{ins}} - \frac{T_w}{L_{ins}} = -mT_0 \tanh(mL_0) + mT_\infty \tanh(mL_0)$$

$$\Rightarrow T_0 = \frac{T_w + T_\infty(L_{ins} \cdot m \cdot \tanh(mL_0))}{1 + L_{ins} \cdot m \cdot \tanh(mL_0)}$$

b) $T_\infty = 20°C = 293.15K$ $T_w = 200°C = 473.15K$

Plugging in the numbers we know:

and $\sqrt{\frac{4h}{kD}} = \sqrt{\frac{4 \cdot 20}{80 \cdot 10 \cdot 10^{-3}}} = 10$

so $$T_0 = \frac{(473.15K) + (293.15K)\left[(0.2m) \cdot 10 \cdot \tanh(10 \cdot 0.2m)\right]}{1 + (0.2m) \cdot 10 \cdot \tanh(10 \cdot 0.2m)}$$

$\Rightarrow T_0 = 361.45K = 81.47°C < T_{max} = 100°C$

Does meet the specified operating condition.

c) So $T_{max} = 100°C = T_0 = 373.15K$ so we get

$$373.15K = \frac{(473.15K) + (293.15K)\left[(0.2m) \cdot 10 \cdot \tanh(10 \cdot L)\right]}{1 + (0.2m) \cdot 10 \cdot \tanh(10 \cdot L)}$$

Proceeding by iteration, $L = 0.1m \rightarrow RHS = 364.49\ K$
$L = 0.05m \rightarrow RHS = 386.7\ K$
$L = 0.07m \rightarrow RHS = 374.64\ K$
$L = 0.071m \rightarrow RHS = 374.18\ K$
$L = 0.072m \rightarrow RHS = 373.73\ K$
$L = 0.073m \rightarrow RHS = 373.29\ K$

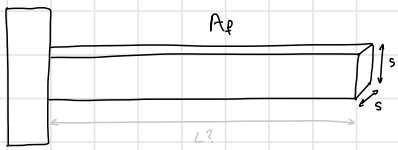
so $L = 0.073m$

3. **KNOWN**: square pin fin, uniform cross-section A_c. $k = 400\,W/m\cdot K$
$s = 10\,mm$ $h = 100\,W/m^2\cdot K$ $\eta_f = 0.6$. Fin has convection tip.

$$-k\frac{dT}{dx}\Big|_{x=L} = h(T(L)-T_\infty)$$

FIND: a) fin length, L b) R_f and ε_f

SCHEMATIC:



ASSUMPTIONS: i) ideal fin ii) heat transfer by conduction is assumed to be 1D. iii) T is uniform within A_f iv) $T = T(x)$

ANALYSIS: a) For a convection tip condition, $q_f = M\dfrac{\sinh(mL)+(h/mk)\cosh(mL)}{\cosh(mL)+(h/mk)\sinh(mL)}$

with $\theta_b = T_b - T_\infty$, $m^2 = \dfrac{hP}{kA_c}$ and $M = \sqrt{hPkA_c}\cdot\theta_b$

$A_c = s^2$
$A_f = 4Ls + s^2$ exact area.
$P = 4s$

We also have $\eta_f = \dfrac{q_f}{q_{max}}$ with $q_{max} = hA_f\theta_b$ surface area of fin

so $\eta_f = \sqrt{hPkA_c}\cdot\cancel{\theta_b}\cdot\dfrac{\sinh(mL)+(h/mk)\cosh(mL)}{\cosh(mL)+(h/mk)\sinh(mL)}\cdot\dfrac{1}{hA_f\cancel{\theta_b}} = \dfrac{\sqrt{h4sks^2}}{h4Ls+hs^2}\,\dfrac{\sinh(mL)+(h/mk)\cosh(mL)}{\cosh(mL)+(h/mk)\sinh(mL)}$

We can plug in numerical values, $m = \sqrt{\dfrac{100\cdot 4\cdot 10\cdot 10^{-3}}{400\cdot(10\cdot 10^{-3})^2}} = \underset{m^{-1}}{10}$ so $\dfrac{h}{mk} = \dfrac{100}{400\cdot 10} = 0.025$

$\sqrt{4hks^3} = \sqrt{4\cdot 100\cdot 400(10\cdot 10^{-3})^3} = 0.4$, $4hs = 4\cdot 100\cdot 10\cdot 10^{-3} = 4$, $hs^2 = 100\cdot(10\cdot 10^{-3})^2 = 0.01$

$$\Rightarrow \quad 0.6 = \frac{0.4}{4L+0.01}\,\frac{\sinh(10L)+0.025\cdot\cosh(10L)}{\cosh(10L)+0.025\cdot\sinh(10L)}$$

We can find L by iteration:

$$0.6 = \frac{0.4}{4L+0.01} \; \frac{\sinh(10L)+0.025\cdot\cosh(10L)}{\cosh(10L)+0.025\cdot\sinh(10L)}$$

Let's start at $L = 0.1m \rightarrow$ using scientific calculator, $RHS = 0.75$

$L = 0.2m \rightarrow RHS = 0.48$

$L = 0.15m \rightarrow RHS = 0.596$

$L = 0.145m \rightarrow RHS = 0.61$

$L = 0.147m \rightarrow RHS = 0.605$

$L = 0.148m \rightarrow RHS = 0.602$

$L = 0.149m \rightarrow RHS = 0.599 \approx 0.6$ so $L = 0.149m$

We could also use the corrected length approximation:

$$\eta_f = \frac{\tanh mL_c}{mL_c} \quad \text{with } m = \sqrt{\frac{hP}{kA_c}} = \sqrt{\frac{h4s}{ks^2}} = 10m^{-1}$$

so $$0.6 = \frac{\tanh(10L_c)}{10L_c}.$$

Similarly by iteration, $L_c = 0.14m \rightarrow 0.63$

$L_c = 0.15m \rightarrow 0.603$

so $$L = L_c - \frac{s}{4} = 0.15 - \frac{10\cdot10^{-3}}{4} = 0.1475m = L$$

b) $$R_f = \frac{1}{\eta_f h A_f} = \frac{1}{\eta_f h\cdot(4Ls+s^2)} = \frac{1}{0.6\cdot100\cdot(4\cdot0.149\cdot10\cdot10^{-3}+(10\cdot10^{-3})^2)}$$

exact

$\Rightarrow R_f = 2.75\ K/W$

and $$\varepsilon_f = \frac{q_f}{q_{nf}} = \frac{q_{max}\eta_f}{q_{nf}} = \frac{\cancel{h}A_f\cancel{\theta_b}\cdot\eta_f}{\cancel{h}A_c\cancel{\theta_b}} = \frac{(4Ls+s^2)\cdot\eta_f}{s^2} = \frac{(4\cdot0.149\cdot10\cdot10^{-3}+(10\cdot10^{-3})^2)\cdot0.6}{(10\cdot10^{-3})^2}$$

$\Rightarrow \varepsilon_f = 36.36$